

KATHMANDU UNIVERSITY
End Semester Examination
March/April, 2017

Marks Scored:

Level : B.E./B.Sc./B.Tech.

Course : MATH 101

Year : I

Semester: I

Exam Roll No. :

Time: 30 mins.

F. M. : 20

Registration No:

Date

APR 04 2017

SECTION "A"

[10Q × 1 = 10 marks]

Fill in the blank space(s) by writing the most appropriate word(s) or symbol(s).

1. If $f(x) = x + \cos x$, then the value of the composite function $(f \circ f)(x)$ at $x = \pi/2$ is
2. The value of the $\lim_{x \rightarrow 1^-} \frac{\sqrt{2x}(x-1)}{|x-1|} = \dots\dots\dots$
3. The average rate of change of the function $f(x) = 2 + \sin x$ on the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is
4. The curve $y = x^2 + 4x - 4$ has horizontal tangent at the point $(x_0, y_0) = \dots\dots\dots$
5. The curve $y = x^3 + bx^2 + cx + d$, where b, c, d are constants, will have a point of inflection at a point $x = 1$ for the value of $b = \dots\dots\dots$
6. The average value of the function $f(x) = |x| - 1$ on the interval $[-1, 1]$ is
7. If $F(x) = \int_0^{\tan x} \frac{dt}{1+t^2}$, then $\frac{dF}{dx} = \dots\dots\dots$
8. The limit of the converging sequence $\{a_n\}_{n=1}^{\infty}$ of real numbers $a_n = \frac{\ln n}{n}$ is
9. The row rank of the matrix $A = \begin{pmatrix} 1 & -1 & 3 \\ 2 & -2 & 6 \end{pmatrix}$ is
10. The set of vectors $\{(0, 2, 1), (-1, 1, k), (1, 0, 3)\}$ in $\mathbb{R}^3$ will be linearly dependent if $k = \dots\dots\dots$

SECTION "B"

[10 Q. $\times$ 1 = 10 marks]

Fill in the blank space(s), DO NOT TICK, by selecting the most appropriate answers from among the given ones.

11. The range of the function $f(x) = \sqrt{1 - x^2}$ is
 [$[-1, 1]$; $[-1, 0]$; $[0, 1]$; Null set]
12. If the graph of the curve $y = \sin x$ is shifted 3 units downward and 2 units left, then the equation of new curve is
 [$y = \sin(x - 2) - 3$; $y = \sin(x + 2) + 3$; $y = \sin(x - 2) + 3$; $y = \sin(x + 2) - 3$]
13. The function $f(x) = \frac{x^2 - 16}{x^2 - 3x - 4}$ has a/an discontinuity at the point $x = -1$.
 [removable; jump; infinite; oscillating]
14. If the surface area of the uniform spherical ice ball of radius R melts at the rate $2 \text{ m}^2/\text{s}$, then the radius of the ball decreases at the rate m/s .
 [$\frac{1}{4\pi R}$; $\frac{1}{2\pi R^2}$; $\frac{1}{2\pi R}$; $\frac{1}{\pi R}$]
15. The function $f(x) = \alpha x^2 - 4x + 1$ will have local minimum value at the point $x = 1$ if value of the constant α is
 [-1 ; 1 ; -2 ; 2]
16. The horizontal asymptote of the curve $y = \frac{2x^2 - 3x + 1}{x^2 - 4}$ is
 [$x + 2 = 0$; $x - 2 = 0$; $y - 2 = 0$; $y + 2 = 0$]
17. If the velocity of a body moving along s -axis at time t seconds is $v = 10t - 3 \text{ meters/second}$, then the displacement of the body over the time interval $t = 1$ to $t = 3$ seconds, is meters.
 [17 ; 34 ; 51 ; 68]
18. The alternating p -series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^p}$ converges conditionally for the value(s) of p satisfying
 [$p = 0$; $p < 0$; $0 < p \leq 1$; $p > 1$]
19. A homogeneous system of linear equations $AX = 0$, where A is $m \times n$ matrix with row rank r and $X \in \mathbb{R}^n$, has unique solution if
 [$m = n$; $m = r$; $n = r$; $n \neq r$]
20. If an eigen value of an invertible matrix A is $\lambda = 2$, then the eigenvalue of the inverse matrix A^{-1} is
 [$\frac{1}{2}$; 2 ; 0 ; $\frac{1}{4}$]

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Year : I
Time : 2 hrs. 30 mins.

Course : MATH 101
Semester : I
F. M. : 55

SECTION "C"

[3 Q. × 7 = 21 marks]

1. State the Mean Value Theorem for the function defined on $[a, b]$ and explain it geometrically. How is this theorem related to the Rolle's Theorem? Also, evaluate the values of α , β and γ such that the function $f(x)$ defined as
- $$f(x) = \begin{cases} -3, & \text{for } x = -1 \\ x^2 - 2\beta x + \alpha, & \text{for } -1 < x < 0 \\ \gamma x + \beta, & \text{for } 0 \leq x \leq 1 \end{cases}$$
- satisfies the hypotheses of the mean values theorem on the interval $[-1, 1]$? [2+1+4]

OR

- What is meant by an indeterminate form of the limit of a function at a point? Also, state the L' Hôpital rule for the limit of a function having the form $\frac{0}{0}$ and use it to evaluate $\lim_{x \rightarrow 0^+} (a^x - b^x)^{1/x}$, where $a > b > 0$ are scalars. [2+2+3]
2. Write geometrical meaning of the definite integral $\int_a^b f(x)dx$ of a function $f(x) \geq 0$ defined on the interval $[a, b]$. Also, find the area of the plane region bounded by the graph of $f(x) = 2x^2 - 3x + 2$, $-1 \leq x \leq 1$ with x -axis using Riemann sum and then verify the answer. [1+4+2]
3. Define a vector space over the real field. Define the linear dependency of the set of vectors $\{u_1, u_2, \dots, u_k\}$, $k \geq 2$ in the vector space $\mathbb{R}^n$. Also, show that the given set of vectors $\{(1, 4, -2); (-2, -7, 4); (-4, -13, 8)\}$ in $\mathbb{R}^3$ is linearly dependent and write down a dependency equation. [2+1+4]

SECTION "D"

[6 Q. × 4 = 24 marks]

4. Evaluate $\frac{dy}{dx}$ (ANY TWO):
(i) $e^{xy} = xy$ (ii) $x = \sin\theta\sqrt{\cos 2\theta}$, $y = \cos\theta\sqrt{\cos 2\theta}$ (iii) $xy + y^2 = 1$ at $(0, -1)$
5. Evaluate the following integrals (ANY TWO):
(i) $\int \frac{1-x}{8+2x-x^2} dx$ (ii) $\int_0^{\pi/2} \frac{d\theta}{2+\cos\theta}$ (iii) $\int_0^\infty xe^{-x^2} dx$
6. Find the interval(s) on which the function $g(x) = x^4 - 4x^3 + 4x^2$, $-4 \leq x \leq 4$ (i) increases and decreases (ii) concave up and concave down. Also, find the point of inflection of $g(x)$.
7. Find the length of the plane curve given by $x = t - \sin t$, $y = 1 - \cos t$, $0 \leq t \leq 2\pi$.

OR

Find the area of the surface generated by revolving the plane curve $x = \frac{y^{\frac{3}{2}}}{3} - y^{\frac{1}{2}}$, $1 \leq y \leq 3$, about y -axis.

8. Use the integral test to prove that the p -series : $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent for $p > 1$.

9. Solve the homogeneous system of linear equations

$x_1 + 3x_2 - 2x_3 + x_4 = 0$; $-3x_1 - 9x_2 + 6x_3 - 4x_4 = 0$; $-x_1 - 3x_2 + 2x_3 - 6x_4 = 0$
by using Gauss-eliminations and back addition to give the parametrically represented solutions. Also, write two non-zero specific solutions.

SECTION "E"

[5 Q. $\times$ 2 = 10 marks]

10. Find eigen values of the matrix $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$.

11. Show that the transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (x - y, x + y)$ is a linear transformation.

12. Solve the differential equation $\frac{dy}{dx} = \frac{x}{x^2+1}$ with initial condition $y(0) = 1$.

13. Find all asymptotes of the curve $y = \frac{x^2 - x + 1}{x - 1}$.

14. Find the quadratic approximation of the function $f(x) = \ln(1 + x)$ at $x = 0$.